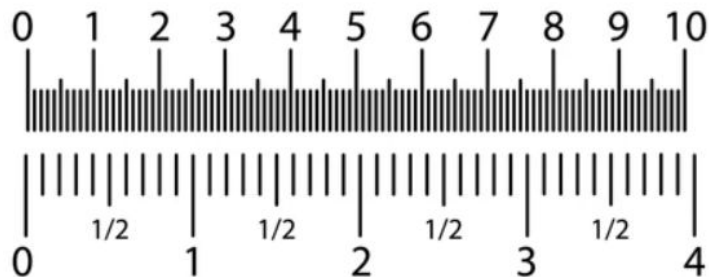




Plain, Diagonal and Vernier Scale

CSE200



Scales



How to draw huge or tiny objects with precision?

Scales



We need to reduce or enlarge the dimensions using some factor

Scales

Scale is a ratio that represents the relationship between the size of something and its representation, such as on a map or a model.

Types of Scale Ratios:

- **Full Scale (1:1)**: Represents drawing is the same size as drawn object.
- **Reduction Scale** (e.g., 1:10, 1:100): Represents drawing is smaller than actual.
- **Enlargement Scale** (e.g., 10:1, 100:1): Represents the drawing is larger than actual.

Scales

The **representative factor (RF)** is a dimensionless ratio used in maps, drawings, and models to represent the relationship between a measured distance on the representation and the corresponding actual distance in reality.

$$\text{RF} = \frac{\text{Distance on the map or drawing}}{\text{Actual distance on the ground or object}}$$

Examples:

1. If a map has an RF of 1:50,000, this means 1 unit on the map (e.g., 1 cm) represents 50,000 units in real life (e.g., 50,000 cm or 500 meters).
2. In architectural drawings, an RF of 1:100 means that 1 cm in the drawing corresponds to 100 cm (1 meter) in reality.

Scales

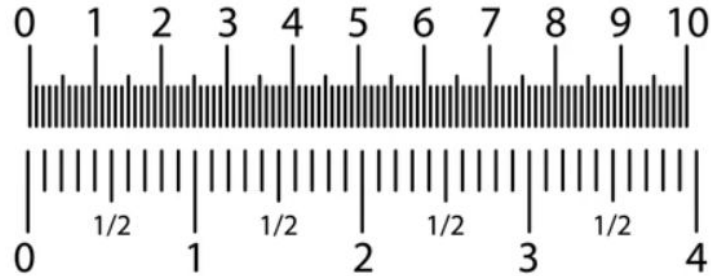
Problem:

If 1.5 meter is represented by a 15 cm scale , what is the RF of that scale?

$$\text{RF} = (\text{Mapped distance}) / (\text{Actual distance})$$

[N.B. 100 cm is equivalent to 1 meter.]

Plain Scale Problem



Plain Scale Problem

Problem: Construct a scale of RF 1:4 to show centimeters and long enough to measure upto 5 decimeters.

$$\text{RF} = 1:4$$

Max Length = 5 decimeters

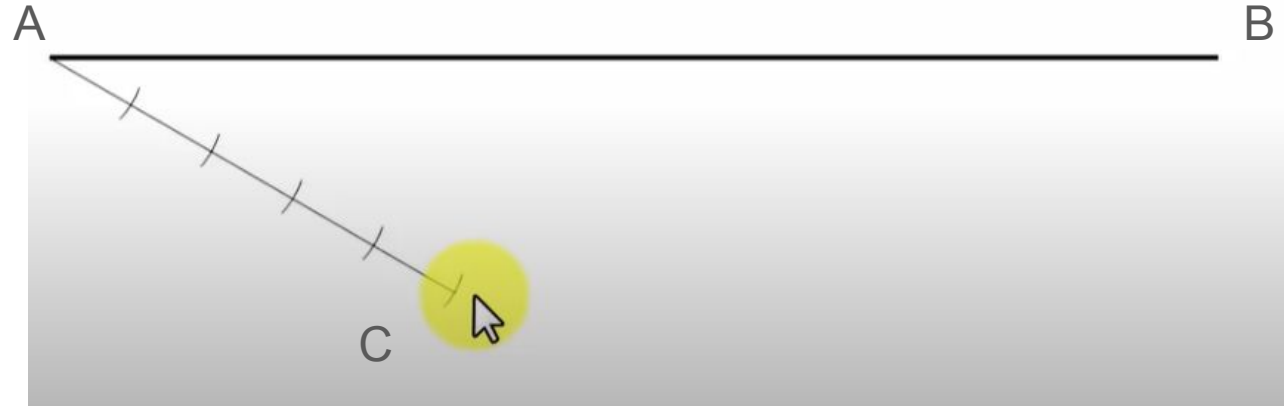
$$\begin{aligned}\text{Length of Scale (in cms)} &= \text{RF (Representative Factor)} \times \text{ML (Maximum Length)} \\ &= \left(\frac{1}{4}\right) \times 5 \text{ dm} \\ &= \left(\frac{1}{4}\right) \times 5 \times 10 \text{ cm} \\ &= 12.5 \text{ cm}\end{aligned}$$

Plain Scale Problem

Draw a line of 12.5 cm.

Draw a line with any angle with the previous line.

Divide that line into 5 segments.

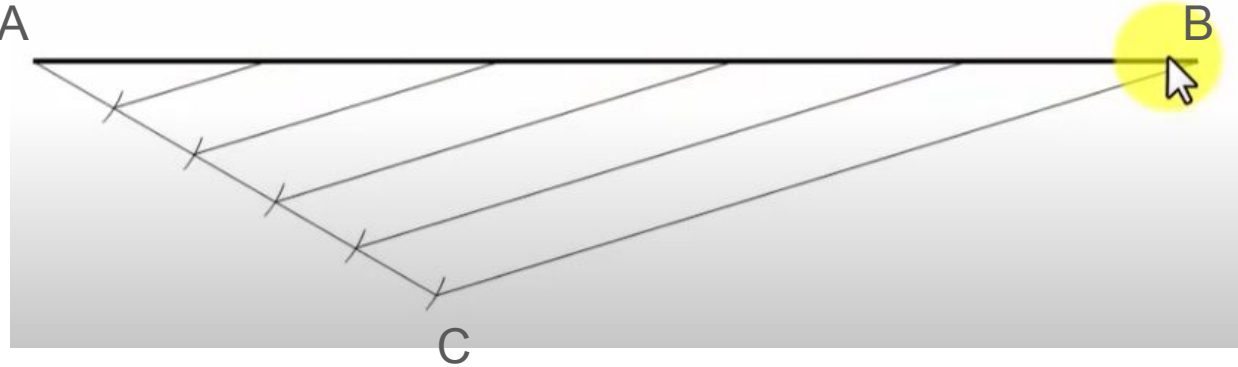


Plain Scale Problem

Join C and B.

Draw 4 lines parallel to BC that will divide the line AB into 5 equal parts.

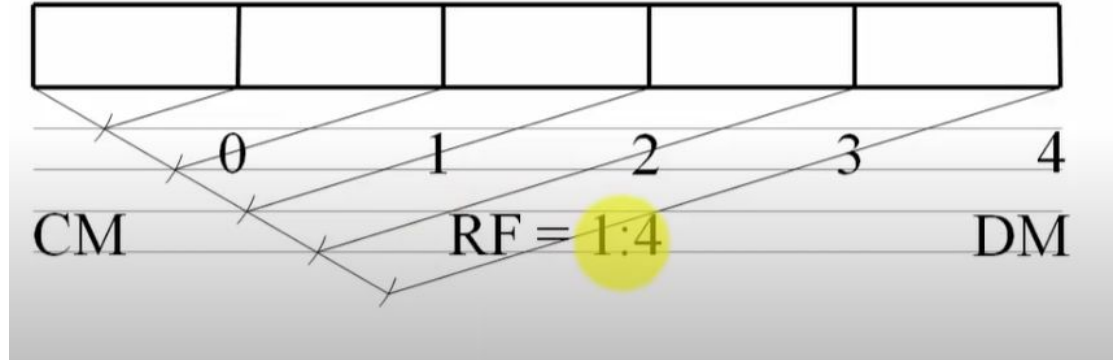
Each part represents 1 dm.



Plain Scale Problem

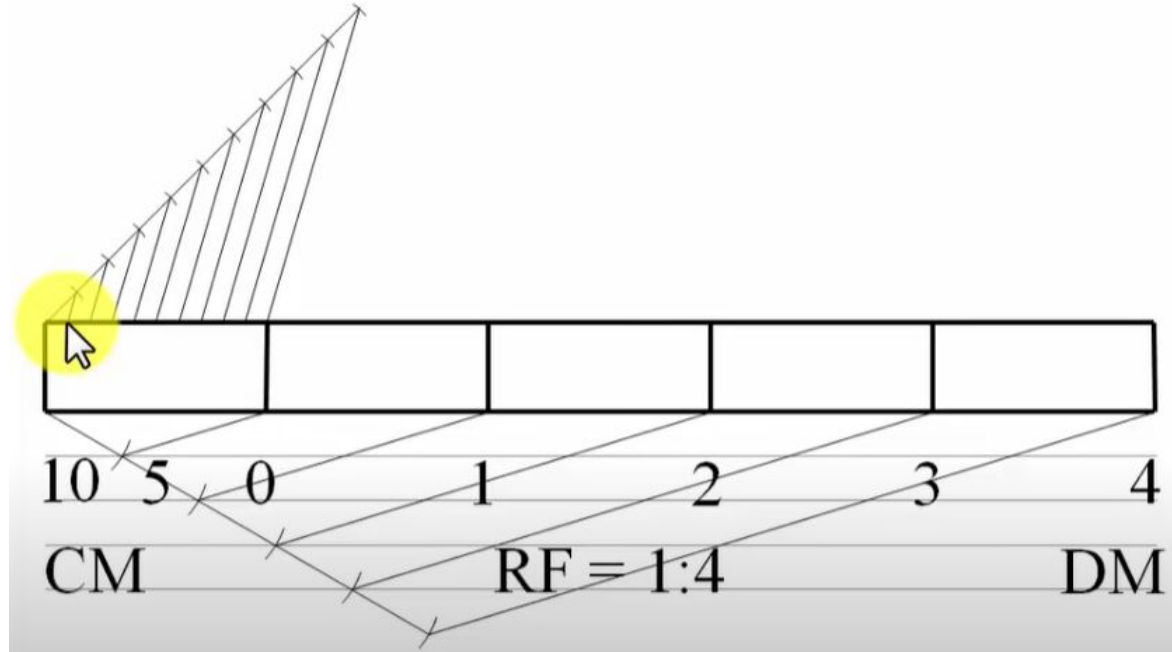
The purpose of these lines are lettering and numbering.

On left we write the true unit, on right we write the mapped unit.



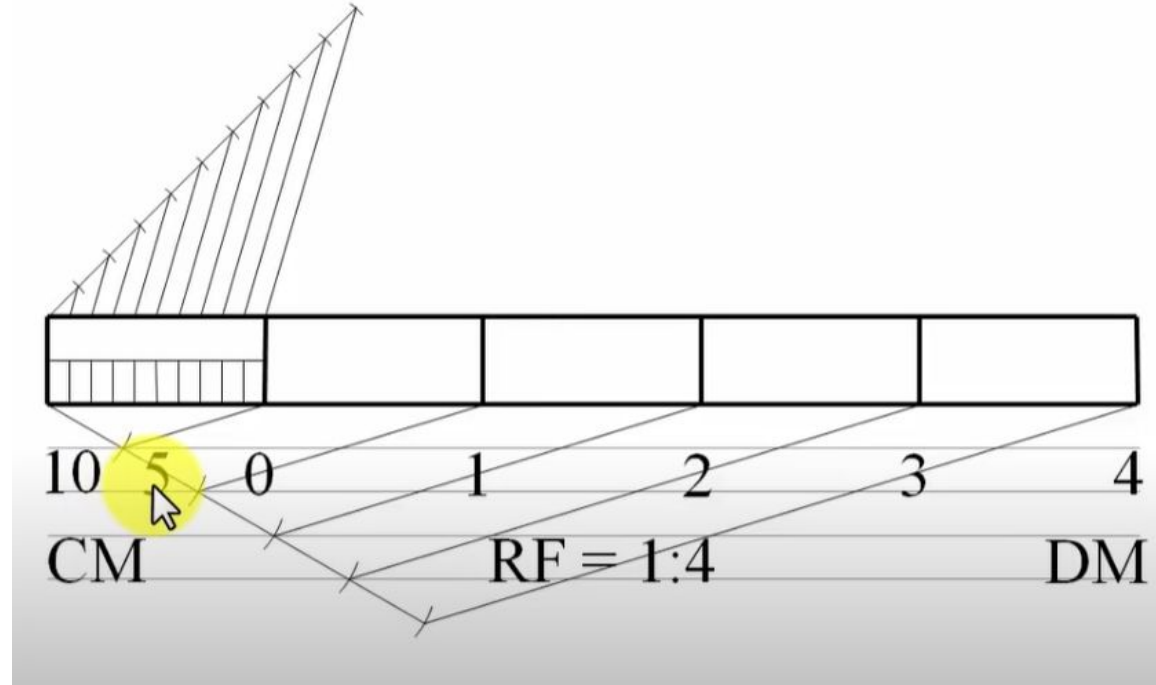
Plain Scale Problem

Similarly we will divide the first segment into 10 equal parts to measure 1 cm.



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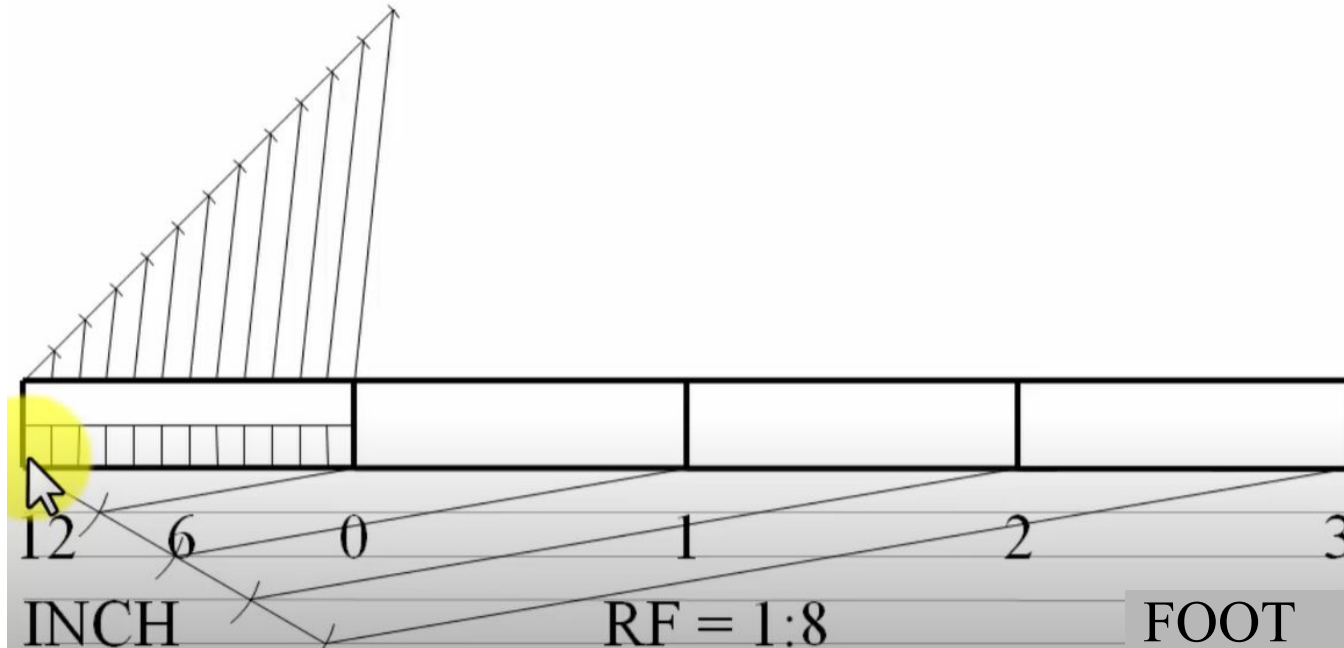


Plain Scale Problem

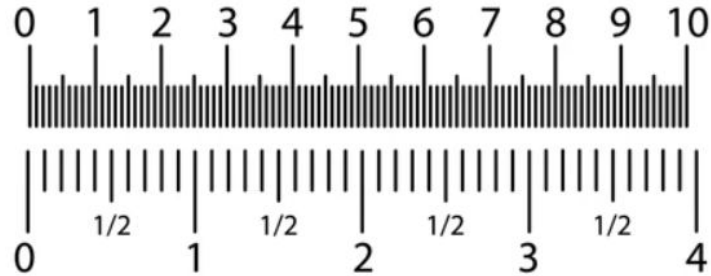
Problem: Construct a scale of 1 foot= 1.5 inch in drawing to read yards and feet and long enough to measure 4 feet.

Plain Scale Problem

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Diagonal Scale Problem



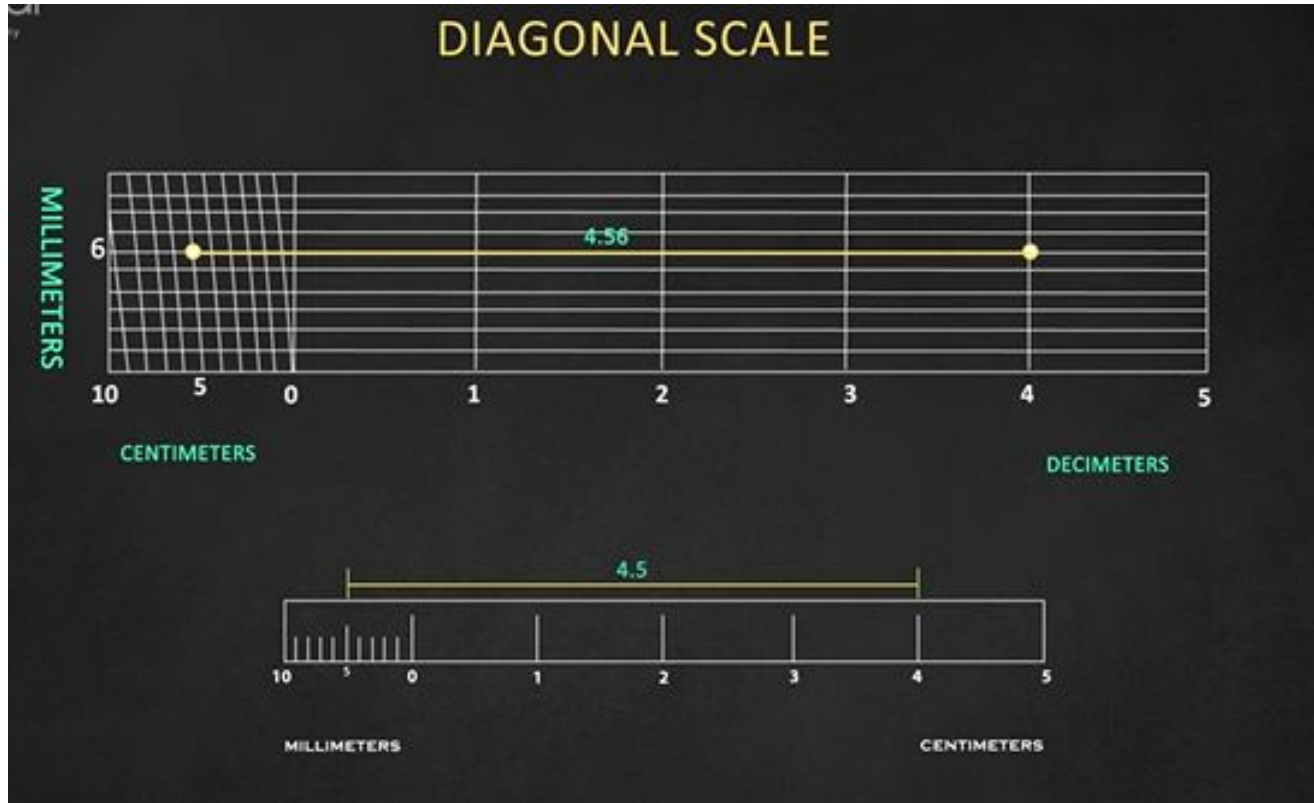
Diagonal Scale Problem

Diagonal Scale is used to represent three consecutive units i.e. a unit and its immediate two sub divisions.

For example

- (a) metre, decimetre and centimetre
- (b) kilometre, hectometre and decametre
- (c) yards, feet and inches etc.

Diagonal Scale Problem



Diagonal Scale Problem

Problem: Construct a diagonal scale of RF= $1/32$ showing yards, feets and inches and to measure upto 4 yards.

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$$LoS = (1/32) \times 4 \text{ yds}$$

$$= (1/32) \times 4 \times 3 \text{ fts}$$

$$= (1/32) \times 4 \times 3 \times 12 \text{ inches}$$

$$= (1/32) \times 4 \times 3 \times 12 \times 2.54 \text{ cm}$$

$$= 11.4 \text{ cm (approximately)}$$

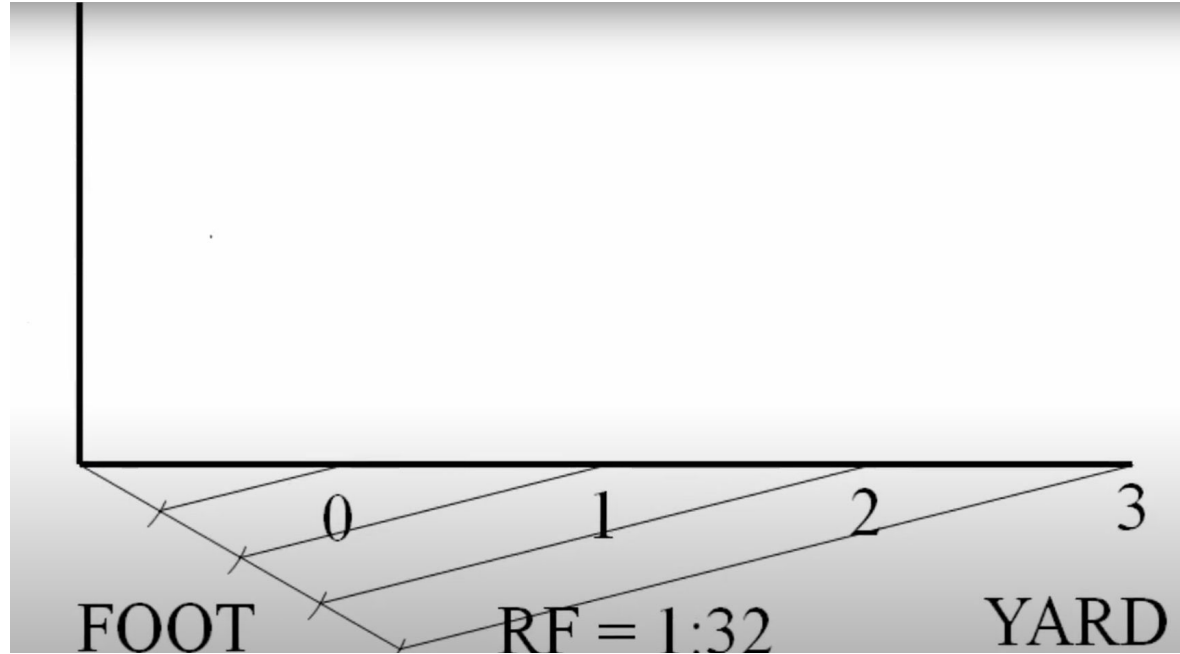
Diagonal Scale Problem

Draw a line of length 11.4 cm.

Divide it into 4 equal parts (in the same way we did in plain scale problem).

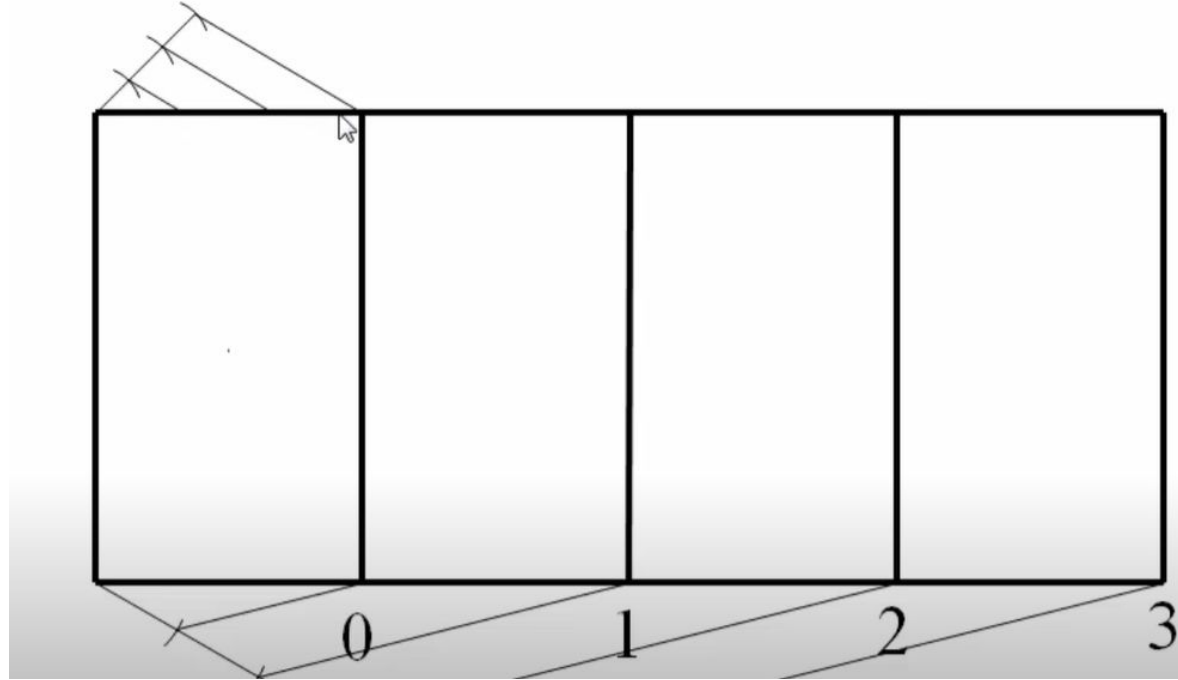
Each segment will represent 1 yard.

Draw a perpendicular line of any (5-7 cm) length



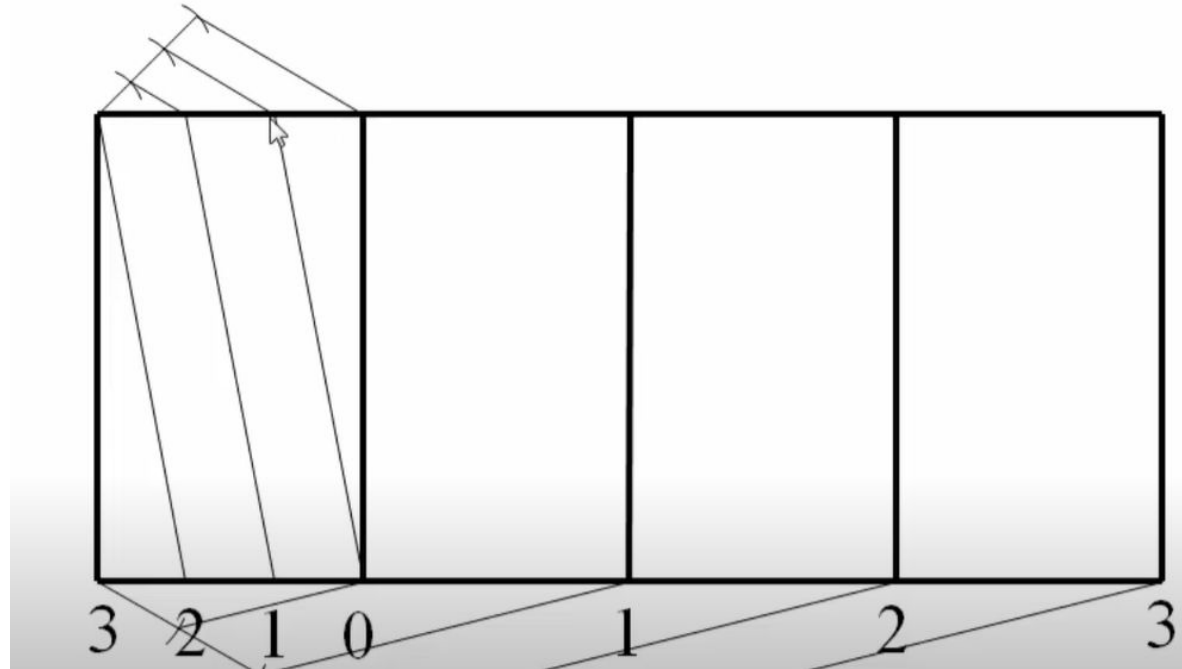
Diagonal Scale Problem

Now we will divide one segment into 3 equal parts (as 3 feet=1 yard).



Diagonal Scale Problem

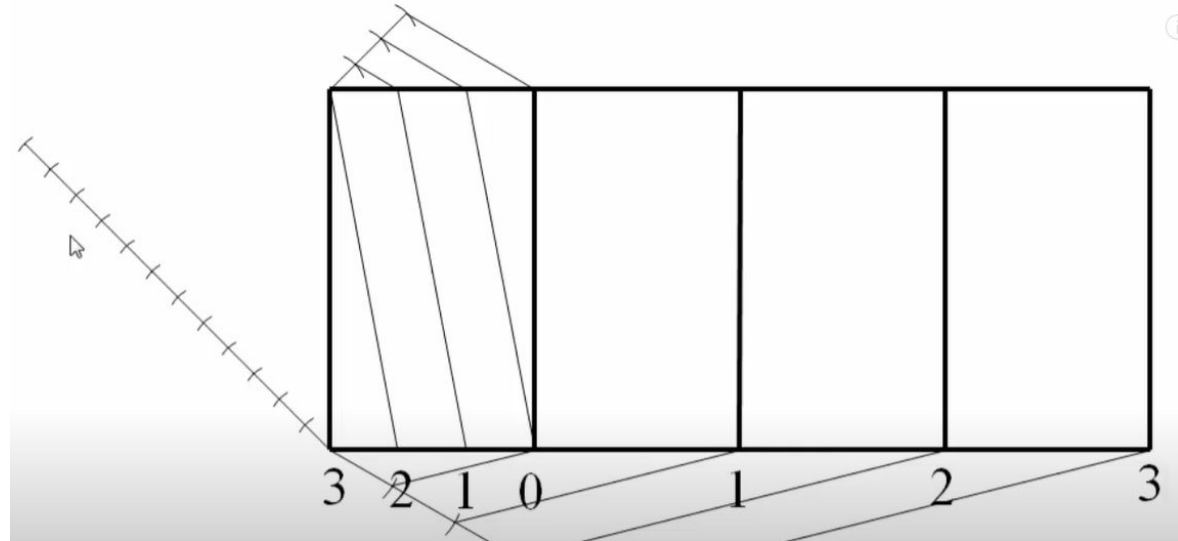
Now we will divide one segment into 3 equal parts (as 3 feet=1 yard).



Diagonal Scale Problem

Now draw another inclined to the left edge.

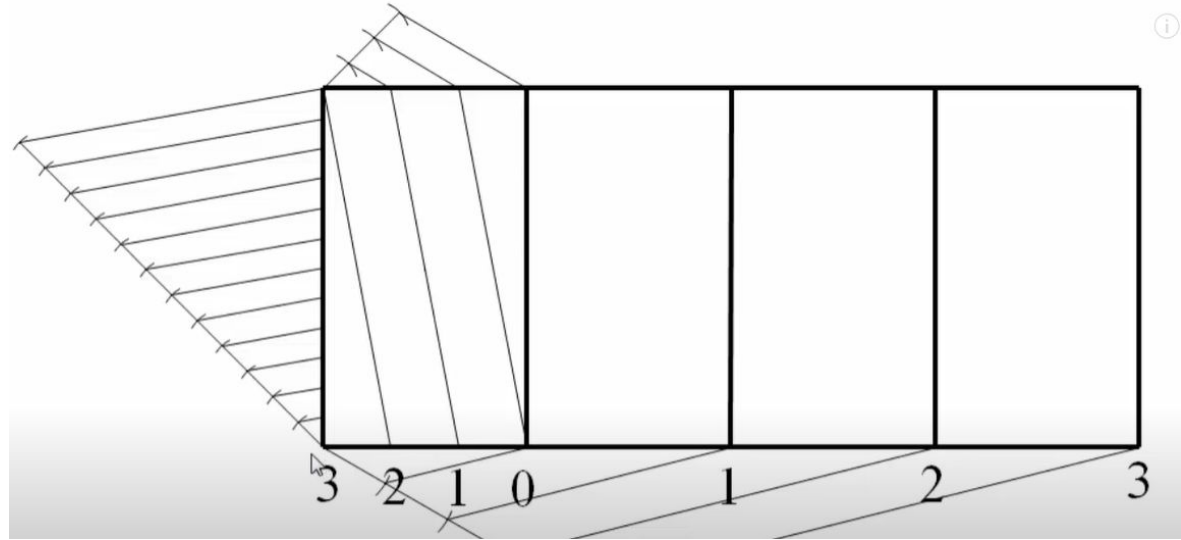
Divide it into 12 equal parts.



Diagonal Scale Problem

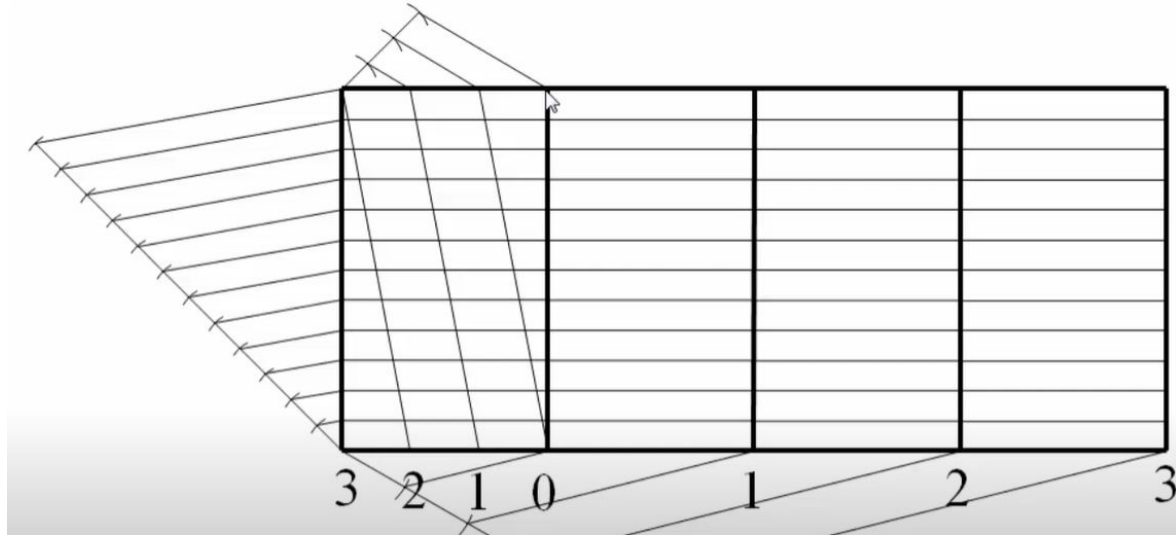
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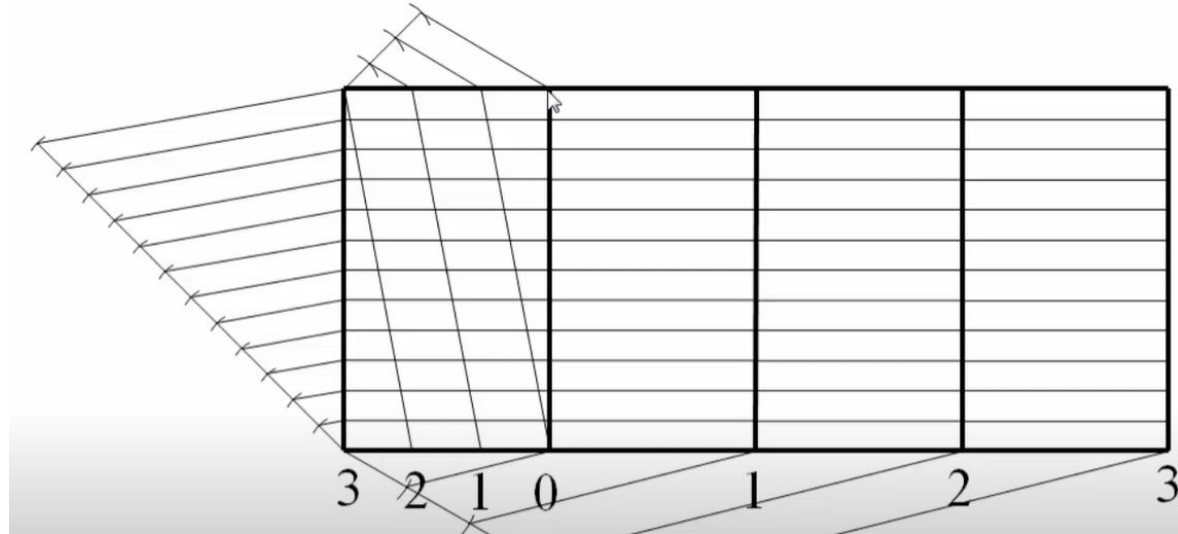
Diagonal Scale Problem

Draw 12 horizontal lines.



Diagonal Scale Problem

Draw 12 horizontal lines.



Diagonal Scale Problem

Our scale is ready to measure inches, foot and yard with an RF of 1:32.

